

DATA CLEANING & REPORTING AUTOMATION

1. INTRODUCTION

This project focuses on automating data cleaning and reporting processes using Python. The main objective of this project is to clean raw data, remove unnecessary information, generate useful insights, and create visual reports automatically.

Data cleaning is an important step in data analysis because real-world datasets often contain duplicate values, missing values, and inconsistent formatting. Automation helps reduce manual work and improves efficiency.

2. OBJECTIVE

The objectives of this project are:

- To understand data preprocessing techniques
- To automate data cleaning using Python
- To remove duplicate records
- To standardize dataset formatting
- To generate visual reports and summaries
- To improve reporting efficiency

3. TOOLS AND TECHNOLOGIES USED

- Python
- Pandas Library
- Matplotlib Library
- Jupyter Notebook / VS Code
- CSV Dataset

4. DATASET USED

Dataset Name: gender_submission.csv

The dataset contains passenger survival information from the Titanic dataset. It includes columns such as:

- PassengerId
- Survived

The dataset was used for cleaning, analysis, and visualization purposes.

5. PROJECT IMPLEMENTATION

Step 1: Importing Libraries

The required Python libraries such as pandas and matplotlib were imported for data processing and visualization.

Step 2: Loading Dataset

The CSV dataset was loaded into Python using pandas.

Step 3: Data Cleaning

The following cleaning operations were performed:

- Removed duplicate rows
- Standardized column names
- Checked dataset structure
- Prepared data for analysis

Step 4: Data Analysis

Basic analysis was performed to calculate:

- Total number of passengers
- Survival count
- Survival percentage

Step 5: Data Visualization

Different charts were generated using matplotlib:

- Bar chart for survival count
- Pie chart for survival percentage

These visualizations helped in understanding the dataset more clearly.

Step 6: Report Generation

An automated text report was generated containing:

- Total rows
- Total columns
- Passenger count

The cleaned dataset was also saved as a new CSV file.

6. OUTPUT

The project successfully generated:

- Cleaned dataset
- Automated report
- Survival analysis charts
- Visual summaries

7. LEARNING OUTCOMES

Through this project, the following concepts were understood:

- Data preprocessing
- Data cleaning automation
- Handling duplicates
- Data visualization
- Automated reporting
- Python libraries for data analysis

8. CONCLUSION

This project successfully automated the data cleaning and reporting workflow using Python. The project improved understanding of preprocessing techniques and reporting automation. It also demonstrated how Python can simplify data analysis tasks efficiently.

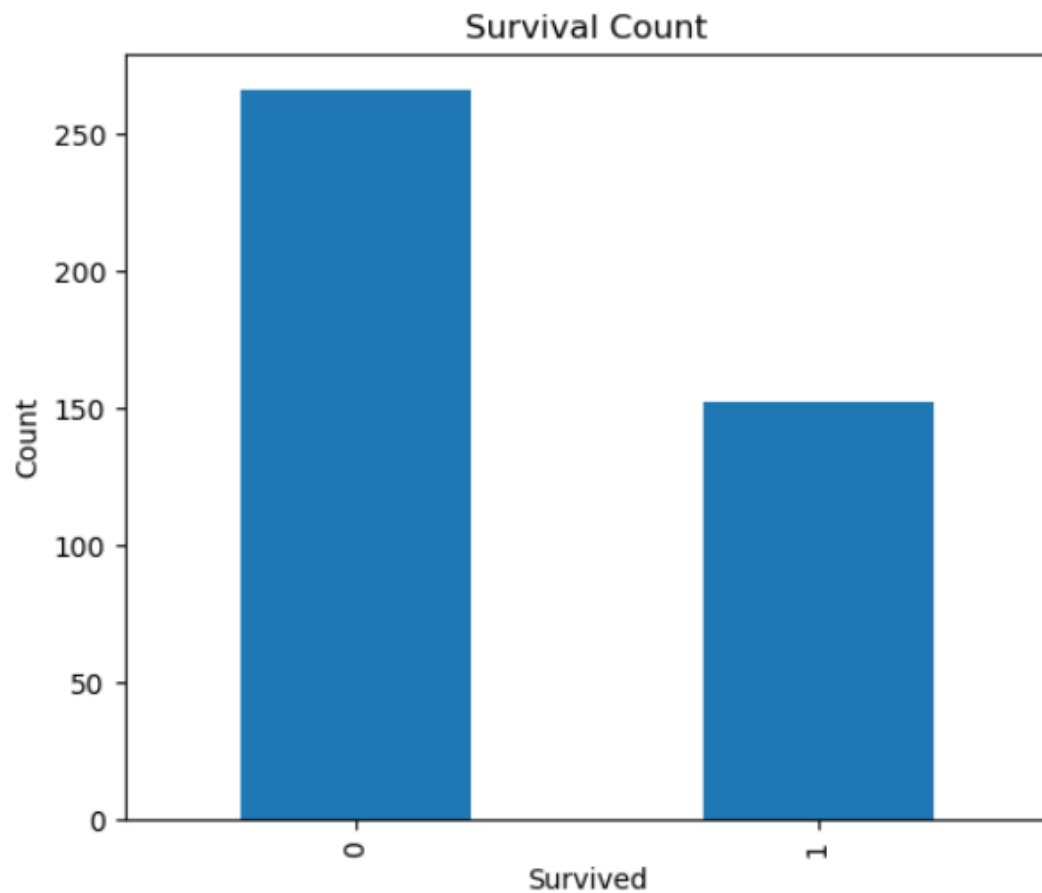
The project achieved all the expected outcomes and provided practical experience in handling real-world datasets.

OUTPUTS :

	PassengerId	Survived
0	892	0
1	893	1
2	894	0
3	895	0
4	896	1

	PassengerId	Survived
count	418.000000	418.000000
mean	1100.500000	0.363636
std	120.810458	0.481622
min	892.000000	0.000000
25%	996.250000	0.000000
50%	1100.500000	0.000000
75%	1204.750000	1.000000
max	1309.000000	1.000000

Total Passengers: 418



Survival Percentage

